SAURAV BHOWMICK

Data Scientist | ML Engineer | Energy Systems Specialist

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Offenburg, Germany

PROFESSIONAL SUMMARY

Data Scientist with **7+ years of combined industry and research experience**, including **5 years at Volkswagen Group** driving RPA automation and process optimization. Holds a **German M.Sc. in Renewable Energy & Data Engineering** from Hochschule Offenburg. Published researcher at **IEEE GPECOM 2026**, **IEEE AUPEC 2025** and **ICED 2025 (Reviewers Choice Award)** in ML-based anomaly detection for industrial power systems. Expertise in Python, TensorFlow, scikit-learn, SQL, and time-series analysis applied to energy and manufacturing domains.

PROFESSIONAL EXPERIENCE

Jun 2026 – Present

Research Assistant – Applied ML for Power Systems

Institute for Sustainable Energy Systems (INES), Hochschule Offenburg, Germany

★ International IEEE Conference Publication (GPECOM 2026)

- Set up and configure power quality measurement devices and data acquisition infrastructure at industrial grid monitoring sites
- Capture and validate high-resolution time-series measurement data (voltage waveforms, power quality parameters), including timestamp validation, UTC handling and systematic data quality checks
- Analyze grid measurement data and develop ML-based anomaly detection and forecasting models for energy grid systems
- Presented peer-reviewed research on **ML benchmarking for industrial power quality anomaly detection** at IEEE GPECOM 2026 in Naples, Italy

Jan 2026 – Present
May 2025 – Aug 2025

Data Scientist & Analyst (Consultant)

Grydion, Berlin, Germany

- Built a web-based grid simulation platform with integrated ML models (TypeScript, Python) compliant with ISO 50001 energy management standards
- Extracted and analyzed large-scale energy system data from MS SQL databases; built Grafana dashboards for real-time voltage behavior monitoring
- Identified system inefficiencies and grid instability risks through analysis of unbalanced currents, phase imbalances, and phase angle shifts

Aug 2025 – May 2026

Guest Researcher

Hochschule Offenburg, Germany

★ International IEEE Conference Presentation (AUPEC 2025)

- Conducted research on robust transient anomaly detection in voltage waveforms using hybrid signal processing and ML techniques
- Presented findings at the **IEEE AUPEC 2025 Conference** in Brisbane, Australia

Sep 2024 – Feb 2025

Master Thesis – Data Science & Power Quality Analytics

Institute for Sustainable Energy Systems, HS Offenburg, Germany

★ Published at ICED 2025 Conference – Reviewers Choice Award

- Developed hybrid anomaly detection methods (cross-correlation, FFT, curve fitting) on 6+ months of time-series data from Janitza UMG 801 power quality sensors
- Trained ensemble ML models (BalancedRandomForest, EasyEnsemble) on highly imbalanced datasets, achieving optimal ROC-AUC performance
- Built LSTM deep learning models (TensorFlow) for voltage behavior forecasting and synthetic voltage profile generation

Jan 2024 – Aug 2024

Research Assistant (HiWi)

Institute for Sustainable Energy Systems, HS Offenburg, Germany

- Automated PLC-MQTT data pipelines for the iFEMA e-mobility project – **reduced manual processing time by 30%**
- Built a logging solution for PLC and MQTT server messages into MySQL databases – **improved data accessibility by 40%**
- Automated Node-RED JSON file generation using Modbus and OPC-UA protocols – **saved 20 hours/week**

Sep 2018 – Sep 2023	**Software Engineer – RPA & Automation** *Volkswagen Group Technology Solutions India Pvt. Ltd., Pune, India* ★ **Best Debutant Award & Best Innovation Award (2019)** • Led end-to-end RPA automation projects across Agile sprint cycles – **reduced manual effort by 30%** and **boosted efficiency by 25%** • Managed application packaging (MSI, InstallShield, Intune) within Scrum workflows – **reduced deployment errors by 40%** • Developed PowerShell automation scripts – **cut operational overhead by 20%** and **error rate by 15%** • Collaborated with cross-functional teams across Germany and India in a global delivery model

EDUCATION

Oct 2023 – Feb 2025	**M.Sc. Renewable Energy and Data Engineering** Hochschule Offenburg, Germany • Focus: Energy Economics, Power Electronics, Energy Storage, Renewable Energy, Operations Research, Energy Data Engineering • Thesis: Analysis and Prediction of Power Quality Anomalies in Power-Electronics-Dominated Industrial Systems
Jul 2014 – Jun 2018	**B.E. Mechanical Engineering** Army Institute of Technology, Pune, India • Focus: Manufacturing, Thermodynamics, Machine Dynamics, Robotics, Mechatronics

TECHNICAL SKILLS

Languages:	Python, SQL, PowerShell, C, Java, TypeScript
ML / AI:	Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, Streamlit, MLflow, DVC
Databases:	MySQL, MS SQL, InfluxDB, PostgreSQL
Automation:	Automation Anywhere (RPA), Node-RED, Docker, Git, Grafana, Azure
Engineering:	MATLAB, Ansys, Abaqus
Protocols:	MQTT, Modbus, OPC-UA
Languages:	English (C2), German (B2), Hindi (Native), Bengali (Native)

KEY PROJECTS

Jun – Aug 2024	**Fossil Hard Coal – Generation Analysis & Classification** *Portfolio Project* • Binary classification of coal generation using EDA, Logistic Regression, KNN, Decision Trees; optimized and deployed Random Forest model with ROC-AUC evaluation
Apr – May 2024	**Energy Data Visualization & Time Series Analysis** *Portfolio Project* • Built ARIMA time-series models to analyze the impact of various energy sources on CO_2 emissions with rolling forecast evaluation
Oct – Dec 2023	**Future Energy Prediction – Regression Modelling** *Portfolio Project* • Analyzed energy data and built regression models to predict key milestones—renewables surpassing 85% share, gas plants dropping below 10%, coal phase-out—plus day-ahead energy price forecasting

AWARDS & LEADERSHIP

Jun 2026	**IEEE GPECOM 2026** – Peer-reviewed publication: ML benchmarking for industrial power quality anomaly detection
Nov 2025	**IEEE AUPEC 2025** – Peer-reviewed publication: transient anomaly detection in voltage waveforms
Aug 2025	**Reviewers Choice Award** – ICED 2025 International Conference
Dec 2019	**Best Debutant Award & Best Innovation Award** – Volkswagen Group
2022 – 2023	**Best General Body Member** – Rotaract Club, Pune
2025	Team Lead & Software Manager – ySpace (Volunteering)
2021 – Present	Active Rotaractor – Rotaract Club Pune & Rotaract Club Freiburg
2018 – 2025	5+ Keynote Speaking engagements (English & German)